

Global Wealth University — SmartBio Database Data Dictionary

This document provides the formal **3NF Normalized Relational Database Schema Data Dictionary** for the **SmartBio Attendance & NUC Compliance Management System**. It is formatted for direct inclusion into **Chapter 4 (System Implementation & Database Design)** of the final-year B.Sc. Software Engineering dissertation.

Entity Summary Table

Table Name	Description	Primary Key	Foreign Key
departments	Academic faculties & departments	department_id	—
academic_sessions	Annual university academic sessions	session_id	—
semesters	First, Second, and Summer semesters	semester_id	ses

roles	System roles (Admin, Lecturer, Class Rep, Student, Kiosk)	role_id	—
users		user_id	rol
biometric_credentials	All system users (Staff, Students, Admins)	credential_id	use
courses	NDPA 2023 compliant hashed biometric templates	course_id	dep sen
course_registrations	Registered departmental courses & credit units	registration_id	stu ses
lecture_sessions	Student enrollment records for courses	session_instance_id	cou
attendance_records	Individual classroom lecture sessions held	attendance_id	ses stu res

audit_logs	Immutable NDPA 2023 security and override audit trail	log_id	act
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Detailed Table Specifications

1. departments

Stores university faculty and departmental classifications.

Column Name	Data Type	Nullable	Key	Default
department_id	INT	No	PK	AUTO_INCREMENT
department_code	VARCHAR(10)	No	UK	—
department_name	VARCHAR(100)	No	—	—
faculty	VARCHAR(100)	No	—	—
created_at	TIMESTAMP	No	—	CURRENT_TIMESTAMP

2. academic_sessions

Maintains university academic calendars.

Column Name	Data Type	Nullable	Key	Default	Description
session_id	INT	No	PK	AUTO_INCREMENT	Unique identifier for academic sessions.
session_name	VARCHAR(20)	No	UK	—	Academic session name, e.g., '2023-2024'.
is_current	BOOLEAN	Yes	—	FALSE	Flag indicating if the session is the current active one.
start_date	DATE	No	—	—	Start date of the academic session.
end_date	DATE	No	—	—	End date of the academic session.

3. semesters

Divides academic sessions into instructional terms.

Column Name	Data Type	Nullable	Key
semester_id	INT	No	PK
session_id	INT	No	FK
semester_type	ENUM('FIRST', 'SECOND', 'SUMMER')	No	—

is_active	BOOLEAN	Yes	—
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4. roles

Implements Role-Based Access Control (RBAC).

Column Name	Data Type
role_id	INT
role_name	ENUM('ADMIN', 'LECTURER', 'CLASS_REP', 'STUDENT')
description	VARCHAR(255)

5. users

Contains profiles for all university actors.

Column Name	Data Type	Nullable	Key	Default
user_id	INT	No	PK	AUTO_INCREMENT
identifier	VARCHAR(50)	No	UK	—
full_name	VARCHAR(150)	No	—	—
email	VARCHAR(100)	No	UK	—
phone_number	VARCHAR(20)	Yes	—	NULL
password_hash	VARCHAR(255)	No	—	—
role_id	INT	No	FK	—

department_id	INT	Yes	FK	NULL
academic_level	INT	Yes	—	NULL
profile_photo_url	VARCHAR(255)	Yes	—	NULL
is_active	BOOLEAN	Yes	—	TRUE
created_at	TIMESTAMP	No	—	CURRENT_TIME

6. biometric_credentials (NDPA 2023 Compliant)

Maintains cryptographic biometric descriptors without storing raw fingerprint images.

Column Name	Data Type
credential_id	INT
user_id	INT
credential_type	ENUM('WEBAUTHN','FINGERPRINT_OPTI
biometric_template_hash	VARCHAR(255)
finger_position	ENUM('RIGHT_THUMB','RIGHT_INDEX',
registered_at	TIMESTAMP
is_revoked	BOOLEAN

7. courses

Maintains departmental course catalogs and regulatory thresholds.

Column Name	Data Type	Nullable	Key	Default
course_id	INT	No	PK	AUTO_INCREMENT
course_code	VARCHAR(15)	No	UK	—
course_title	VARCHAR(150)	No	—	—
credit_units	INT	No	—	3
department_id	INT	No	FK	—
level	INT	No	—	—
semester_id	INT	No	FK	—
lecturer_id	INT	Yes	FK	NULL
minimum_attendance_pct	DECIMAL(5,2)	No	—	75.00

8. course_registrations

Maps enrolled students to their registered courses.

Column Name	Data Type	Nullable	Key	Default
registration_id	INT	No	PK	AUTO_INCREMENT
student_id	INT	No	FK	—
course_id	INT	No	FK	—
session_id	INT	No	FK	—
registration_date	TIMESTAMP	No	—	CURRENT_TIMESTAMP

is_approved	BOOLEAN	Yes	—	TRUE
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9. lecture_sessions

Records each instance of a lecture delivered.

Column Name	Data Type
session_instance_id	INT
course_id	INT
lecturer_id	INT
topic	VARCHAR(200)
venue_hall	VARCHAR(100)
start_time	TIMESTAMP
end_time	TIMESTAMP
status	ENUM('ACTIVE', 'CONCLUDED', 'CANCELLED')

10. attendance_records

Captures biometric attendance events and exception resolutions.

Column Name	Data Type
attendance_id	INT
session_instance_id	INT
student_id	INT

verification_time	TIMESTAMP
verification_method	ENUM('WEBAUTHN_BIOMETRIC','OPTICA
match_confidence_score	DECIMAL(5,2)
status	ENUM('PRESENT','FLAGGED','ABSENT'
flag_reason	VARCHAR(255)
resolved_by_lecturer_id	INT
resolution_notes	TEXT

11. audit_logs

Provides non-repudiation and NDPA 2023 regulatory compliance.

Column Name	Data Type
log_id	INT
actor_user_id	INT
action_type	ENUM('LOGIN','SESSION_START','SESSION_END'
target_entity	VARCHAR(50)
target_id	VARCHAR(50)
action_details	TEXT
ip_address	VARCHAR(45)
timestamp	TIMESTAMP